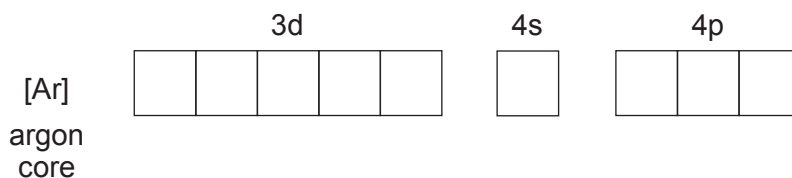


SECTION A

Answer **all** questions in the spaces provided.

1. Using the convention of arrows to represent electrons, complete the electronic structure of an arsenic atom, As. [1]



2. Give the oxidation number of molybdenum in the ion MoO_4^{2-} . [1]

.....

3. The half-life of the radioactive isotope oxygen-15 is 2 minutes.

- (a) Give the atomic number, mass number and symbol of the nuclide produced when 1 atom of ^{15}O decays by the emission of one positron. [1]

.....

- (b) If a quantity of ^{15}O decays by positron emission, determine what fraction of ^{15}O is left after 10 minutes. [1]

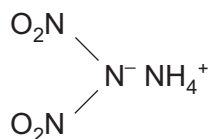
.....

4. Explain why the second molar ionisation energy of sodium is greater than its first molar ionisation energy. [1]

.....
.....
.....

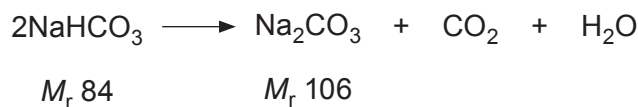


5. The explosive ADN has the structural formula shown below.



Give the empirical formula of ADN. [1]

6. Sodium carbonate, Na_2CO_3 , is made by heating sodium hydrogencarbonate, NaHCO_3 .

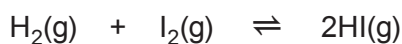


Calculate the atom economy of this reaction. [1]

Atom economy = %

2410U101
03

7. Hydrogen and iodine react together to give hydrogen iodide.



- (a) Write the expression for the equilibrium constant in terms of concentration, K_c , for this reaction. [1]

- (b) The equilibrium constant, K_c , for this reaction has a value of 46.0 at a certain temperature. Calculate the equilibrium concentration of hydrogen at this temperature. The equilibrium concentration of iodine is 1.20 mol dm^{-3} and the equilibrium concentration of hydrogen iodide is $15.00 \text{ mol dm}^{-3}$. [2]

Concentration of hydrogen = mol dm^{-3}

10

